📘 CARDIOVASCULAR DISORDERS IN PREGNANCY

1. BASIC CONCEPT

- Pregnancy puts a **gradual but significant load on the heart** because the body needs to supply blood to both mother and fetus.
- A normal heart adapts to this increased demand without issues.
- A diseased heart fails to adapt, leading to complications like **heart failure, arrhythmias, or even death**.

IMPORTANT

- Cardiovascular disease is responsible for **about 25% of maternal deaths**.
- It affects **1–4% of pregnancies**, but risk is increasing due to lifestyle diseases and delayed pregnancies.

2. PHYSIOLOGICAL CHANGES IN PREGNANCY

2.1 Overall Idea

- Pregnancy creates a **high-volume, low-resistance circulation system**.
- This means more blood is pumped, but the resistance in vessels is reduced to compensate.

2.2 Hemodynamic Changes (Table Explained Properly)

- **Cardiac output increases by about 40–43%**, which is the most important change in pregnancy.
- This increase happens because:
  - Stroke volume increases early in pregnancy.
  - Heart rate increases later in pregnancy.
- **Heart rate increases by around 15–20%**, making the heart beat faster to meet demand.
- **Systemic vascular resistance decreases significantly**, which reduces the workload on the heart.
- **Pulmonary vascular resistance also decreases**, allowing easier blood flow through lungs.
- **Blood pressure slightly decreases in early pregnancy**, then returns to near normal.
- **Colloid osmotic pressure decreases**, which explains why edema (swelling) is common.

2.3 Timeline of Changes

- Cardiac output starts increasing early and rises rapidly by **8 weeks of pregnancy**.
- It reaches its **maximum in mid-pregnancy**.
- The load remains high throughout late pregnancy.
- After delivery, sudden fluid shifts make the **postpartum period the most dangerous**.

2.4 Mechanism (Step-by-Step Logic)

- Hormonal changes cause **vasodilation (widening of blood vessels)**.
- This reduces vascular resistance.
- Reduced resistance increases venous return to the heart.
- Increased venous return increases stroke volume.
- Increased stroke volume increases cardiac output.

👉 This entire process is an **adaptive mechanism**, not a disease.

2.5 Ventricular Changes

- The heart increases in size to accommodate increased blood volume.
- The left ventricle becomes slightly enlarged and more spherical in shape.
- Left ventricular mass increases by about **30–35%** near term.
- Despite these changes, **contractility and ejection fraction remain normal**.

2.6 Clinical Meaning

IMPORTANT

- Pregnancy does not cause a weak or hyperactive heart.
- It causes a **well-adapted high-output state**.
- However, if the heart is already diseased, it cannot handle this load and fails.

3. NORMAL CARDIAC FINDINGS IN PREGNANCY (DIAGRAM EXPLAINED)

The diagram (page 6) shows findings that are **normal but may look abnormal clinically**.

- Mild distension of neck veins can be seen due to increased blood volume.
- Flow murmurs are common due to increased blood flow across valves.
- The second heart sound may appear split or louder.
- A third heart sound (S3) may occasionally be heard.
- Venous hum and mammary souffle (breast-related murmur) can be present.

IMPORTANT

- These findings are **physiological and should not be mistaken for heart disease**.
- The mistake is diagnosing disease based on these normal changes.

4. WHEN TO SUSPECT HEART DISEASE

4.1 Symptoms That Indicate Disease

- Breathlessness that is progressive or occurs at rest is abnormal.

- Difficulty breathing while lying down (orthopnea) suggests cardiac failure.
- Chest pain may indicate ischemic heart disease.
- Coughing blood (hemoptysis) is a serious sign.
- Syncope (fainting) suggests severe cardiac compromise.

4.2 Clinical Signs That Suggest Disease

- Cyanosis indicates poor oxygenation and serious cardiac disease.
- Clubbing suggests chronic heart disease.
- Persistent tachycardia indicates abnormal cardiac stress.
- Loud systolic murmurs (> grade 3) are abnormal.
- Any diastolic murmur is always pathological.
- Cardiomegaly indicates structural heart disease.
- Persistent neck vein distension suggests heart failure.

IMPORTANT RULE

- Mild symptoms = often normal
- Severe or progressive symptoms = **suspect disease**

5. DIAGNOSTIC STUDIES

5.1 ECG (Diagram Explained)

- Heart rate is increased due to physiological demand.
- The heart shifts position upward, causing **left axis deviation**.
- PR interval may be slightly reduced.
- Occasional premature beats are common.

IMPORTANT

- These changes are normal in pregnancy and should not be misinterpreted.

5.2 Chest X-ray

- Safe if proper shielding is used.
- The heart may appear enlarged due to normal anatomical changes.
- Mild enlargement should not be overdiagnosed as disease.

5.3 Echocardiography

- This is the **most important and safest diagnostic tool**.
- It helps evaluate:
    - Cardiac structure
    - Valve function
    - Ventricular function
- Normal findings include:
    - Slight increase in chamber size
    - Mild increase in left ventricular mass
    - Mild valve regurgitation

IMPORTANT

- Ejection fraction remains normal.

5.4 MRI

- Useful in complex cases such as congenital heart disease.
- Provides better structural detail than echocardiography.

5.5 Stress Testing

- Helps assess how well the heart handles physical activity.
- Can detect exercise-induced arrhythmias.

5.6 Invasive Procedures

- Cardiac catheterization can be performed if necessary.
- Radiation exposure to fetus is minimal and usually acceptable.

6. FUNCTIONAL CLASSIFICATION (NYHA)

- This classification is based on how much physical activity a patient can tolerate.
- Class I means no limitation in activity and no symptoms.
- Class II means symptoms occur with normal activity but not at rest.
- Class III means symptoms occur with minimal activity.
- Class IV means symptoms occur even at rest.

IMPORTANT

- Higher class = higher maternal risk.
- Class III and IV patients require close monitoring.

7. WHO RISK CLASSIFICATION (TABLE EXPLAINED)

- WHO classification predicts **risk during pregnancy**.
- WHO 1 indicates no increased risk compared to normal population.
- WHO 2 indicates mild increase in risk.
- WHO 3 indicates significant risk and requires specialized care.
- WHO 4 indicates extremely high risk and pregnancy is contraindicated.

IMPORTANT

- WHO 4 conditions include:
    - Pulmonary hypertension
    - Severe ventricular dysfunction
    - Severe aortic disease

👉 These patients should **avoid pregnancy**.

8. PRECONCEPTION COUNSELING

- Women with heart disease should be evaluated before pregnancy.
- The goal is to optimize cardiac function and reduce complications.
- A multidisciplinary team should be involved, including:
    - Obstetrician
    - Cardiologist
    - Anesthesiologist
- Some heart conditions can be corrected surgically before pregnancy.
- Genetic risk of congenital heart disease should be discussed.

9. RISK FACTORS

- Advanced maternal age (>40 years) increases risk.
- Hypertension significantly contributes to complications.
- Obesity increases cardiac workload and risk.
- Poor access to prenatal care worsens outcomes.
- Socioeconomic factors also play a role.

10. ANTEPARTUM CARE

- Regular monitoring is essential to detect early complications.

- Prevention of infections is important because infections can trigger heart failure.
- Recommended:
    - Influenza vaccination
    - Pneumococcal vaccination
- Patients should avoid:
    - Smoking
    - Recreational drugs

11. LABOR AND DELIVERY

11.1 Preferred Method

- Vaginal delivery is preferred because it causes less stress compared to surgery.
- Cesarean section is reserved for specific cardiac or obstetric indications.

11.2 Indications for Cesarean Section

- Severe aortic disease or aneurysm
- Acute heart failure
- Recent myocardial infarction
- Severe aortic stenosis
- Warfarin use close to delivery

11.3 Monitoring During Labor

- The mother should be kept in a semi-recumbent position.
- Vital signs should be monitored frequently.

Warning signs include:

- Heart rate above 100 bpm
- Respiratory rate above 24/min
- Increasing breathlessness

12. ANALGESIA AND ANESTHESIA

- Pain relief is important to reduce stress on the heart.
- Epidural analgesia is preferred as it provides stable pain control.
- However, epidural can cause hypotension, which must be managed carefully.
- Special care is needed in:

    - Pulmonary hypertension
    - Aortic stenosis

13. INTRAPARTUM HEART FAILURE

- Labor increases cardiac workload due to:
    - Pain-induced catecholamine release
    - Fluid shifts
    - Increased pressure during pushing
- Conditions like preeclampsia, anemia, hemorrhage, and infection can trigger heart failure.

14. POSTPARTUM PERIOD (PUERPERIUM)

MOST CRITICAL PHASE

- After delivery, blood from the uterus returns to circulation suddenly.
- This increases cardiac load rapidly and may lead to heart failure.
- Complications include:
    - Pulmonary edema
    - Thromboembolism
    - Infection
- Continuous monitoring is required even after delivery.

15. FINAL FLOW (CONNECT EVERYTHING)

- Pregnancy increases blood volume and cardiac output.
- The heart adapts through structural and functional changes.
- If the heart is healthy → no problem.
- If diseased → failure occurs.
- Risk is highest during:
    - Late pregnancy
    - Labor
    - Postpartum

16. QUICK REVISION POINTS

- Cardiac output increases by about 40%.
- Vascular resistance decreases significantly.
- Echocardiography is the best diagnostic tool.

- NYHA class tells severity.
- WHO classification tells pregnancy risk.
- Vaginal delivery is preferred.
- Postpartum period is the most dangerous.